# Proof of Construction of $L_2$ Subspace Embedding via JLT matrices

(1) **Claim:** The size of $\gamma$-net of set $S^{d-1}$ is $\leq \left(1+\frac{2}{\gamma}\right)^d$

**Proof:**

Let us construct a $\gamma$-net. Maximally place the points on $S^{d-1}$ s.t no two points are within $\gamma$-distance.

↳ This is a $\gamma$-net

Consider balls of radius $\gamma/2$ around these points. The balls are disjoint. total volume of the balls in

$$|N| \cdot V_d \cdot \left(\frac{\gamma}{2}\right)^d$$

Consider a ball that at center of $S^{d-1}$ of radius $\left(1+\frac{\gamma}{2}\right)$.

This ball encapsulates all balls

Thus

$$|N| \cdot V_d \cdot \left(\frac{\gamma}{2}\right)^d \leq \left(1+\frac{\gamma}{2}\right)^d \cdot V_d$$

$$\boxed{|N| \leq \left(1+\frac{2}{\gamma}\right)^d}$$

(2) Consider a ~~set~~ matrix $A \in R^{n \times d}$.

Consider

$$\$^{(A)} = \{ y \mid \|y\|_2 = 1 \quad y \in CS(A) \}$$

Q. What is ^size of a $\gamma$-net of $S$?

$y = Ux$, $x \in S^{d-1}$

$\exists\ x' \in N(S^{d-1})$

$\|x' - x\|_2 < \gamma$

$\Rightarrow \|Ux' - Ux\|_2 < \gamma$

$\{ Ux \mid x \in N(S^{d-1}) \}$

$\gamma$-net of $\$^{(A)}$.

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(3) Let $S \in R^{k \times d}$. ~~s.t~~ JLT$(\varepsilon, \delta, S^d)$,

$\langle Sw, Sw' \rangle = \langle w, w' \rangle \pm \varepsilon$

$\forall\ w, w'$ ~~in~~ $\frac{1}{2}$-net

~~CS~~ $\$$

then $\|SAx\|_2^2 = (1 \pm \varepsilon)\|Ax\|_2^2$

↳ If we prove this for $x$ s.t $\|Ax\|_2 = 1$ then it is proved for all $x$.



→ $Cs(A) \cap \{Ax \mid \|Ax\|_2 = 1\}$

$\gamma$-net of this

$\gamma = 1/2$

To prove: $\|Sy\|_2^2 = 1 \pm \varepsilon$ $\quad y = Ax$, $\|y\| = 1$

How to proceed.

- Should write $y$ in terms of $w_0 \in$ $\gamma$-net

How:

$y = w_0 + \Delta_0 \qquad w_0 \in \frac{\gamma}{2}\text{-net}$

$\|\Delta_0\|_2 \leq \frac{1}{2}$

$\frac{\Delta_0}{\|\Delta_0\|} = w_1 + \Delta_1 \qquad w_1 \in \frac{\gamma}{2}\text{-net}$

$\frac{\Delta_1}{\|\Delta_1\|} = w_2 + \Delta_2$

$\frac{\Delta_i}{\|\Delta_i\|} = w_i + \Delta_i$

$$y = w_0 + \|\Delta_0\| \cdot w_1 + \|\Delta_0\| \|\Delta_1\|^{1/2} w_2 + \dots$$

$$y = w_0 + \lambda_1 w_1 + \lambda_2 w_2 + \dots$$

$$\lambda_i \leq \frac{1}{2^i} \quad \text{if } w_i \in \tfrac{1}{2}\text{-net}$$

$$\|Sy\|^2 = \langle Sy, Sy \rangle$$

$$= \left\langle S\left(\sum \lambda_i w_i\right), S\left(\sum \lambda_i w_i\right) \right\rangle$$

$$= \left\langle \sum_{i,j} \langle S\lambda_i w_i, S\lambda_j w_j \rangle \right.$$

$$= \lambda_i \lambda_j \sum_{ij} \langle S w_i, S w_j \rangle$$

$$\left(3 \pm \langle w_i, w_j \rangle\right) \sum_{ij} \lambda_i \lambda_j$$

$$= \sum_{ij} \lambda_i \lambda_j \pm \underbrace{\langle w_i, w_j \rangle}_{ij}$$

$$\left(\sum \lambda_i\right)^2 3 \pm \langle y, y \rangle$$

$$\|y\|^2$$

$$\left(\sum \lambda_i\right)^2 3 \pm 1 =$$

$$3 \pm 4$$

Ft:

Choose $JLT\left(\frac{\varepsilon}{4}, \delta, 5^d\right)$

∴ This completes our proof.

→ Hash functions:

$S : k \times d$ Count Sketch Matrix

$k = O\left(\frac{d^2}{\varepsilon^2 \delta}\right)$ only 1 non zero per column is also a $l_2$ - Subspace Embedding

Q: How many random bits are needed to get a Count sketch matrix?

Completely random bits / Multiply Independent

⇒

$S = m \times n$ matrix

$n \cdot \log_2(m)$ random bits

Limited independence

* $X_1 \dots X_n$ are k-wise independent if

~~$X_1 \dots X_n$~~

$\forall$ any $X_{i_1} \dots X_{i_k}$ and vals $y_1 \dots y_k$

$$Pr\left( X_{i_1} = y_1 \cap X_{i_2} = y_2 \dots \right)$$

$$= Pr\left( X_{i_1} = y_1 \right) \cdot Pr\left( X_{i_2} = y_2 \right) \dots$$

How many random bits are necessary to create pairwise ~~random~~ independent bits

2-wise

→ $n$ random bits $\Rightarrow$ pairwise bits?

$\{X_1, \dots, X_m\}$

$\{0\}^{1/2}$
$\{1\}^{1/2}$

$S \subseteq 2^m \qquad S \neq \emptyset$

$|S| \leq 2^{n-1}$

$Y_i = \bigoplus_{j \in S_i} X_j$

$2^{n-1}$ bits

Pairwise Indep.

$$Pr(Y_i = y \cap Y_j = y') = \frac{1}{4}$$

$\oplus_{S_i} X_{j_k}$ $\quad$ $\oplus_{S_j} X_{i_k}$ $\quad$ $\exists \cdot X_i$

$$Pr(Y_i = y \mid Y_j = y')$$

$X_i \in S_i - S_j$ or $S_j - S_i$

Keep everything fixed except $X_i \in S_i - S_j$ WLOG

then the value of the $X_i$ bem 0 or 1 determine the value of $Y_i$

Thus, by

$$Pr(X_i = y \mid Y_j = y') = 1/2$$

$\{X_{i_1} \cdots X_{i_{k_1}}\}$ $\{X_{j_1} \cdots X_{j_{k_2}}\}$

Page No.:
Date:

Create a pairwise random number using two random inputs.

p = prime

$X_1, X_2$ = Units $\{0, \ldots p-1\}$

$$Y_i = (X_1 + i \cdot X_2) \% p$$

→ we get $(p-1)$ random numbers that are pair wise independent

$$Pr(Y_i = y_1 \cap Y_j = y_2) = Pr(Y_i = y_1) Pr($$

$$Pr(Y = y_i) = \frac{1}{p}$$ Why?

~~Hash function~~ :-

$$y_1 = (X_1 + 1 \cdot X_2) \% p$$

$$y_2 = (X_1 + 2 X_2) \% p$$